

INFOMDW: Data Wrangling

Data Extraction (SQL)

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Outlines

- So Far
 - Data types
 - Structured, semi-structured and unstructured
 - Data Models
 - Relational model and entity-relationship model
 - Graphs and trees
 - Objects
 - Data model components
 - Data, structure, semantics, and operations
 - Databases vs file systems
 - DDL and DML
 - Creating and modifying tables (relations) in SQL



Outlines

- Today
 - Reminder
 - Office hours only by appointment (send email)
 - Operations on databases
 - Relational algebra
 - Data Extraction with SQL



Reading Material for Today

Database System Concepts (7th Edition)

CH 2.6, 3.3 – 3.8.



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Relational Algebra



Relational Algebra Operators

- Union $\cup$, intersection $\cap$, difference $-$
- Selection σ
- Projection π
- Join $\bowtie$
- Rename ρ
- Duplicate elimination δ
- Grouping and aggregation γ
- Sorting τ

RA

Remember: a
relation is a **set**
of records

Extended RA



Union

$r \cup s$

r

A	B
α	3
β	2

s

A	B
α	1
β	2
γ	3

$r \cup s$

A	B
α	1
β	2
γ	3
α	3



Intersection

$$r \cap s$$

r

A	B
α	3
β	2

s

A	B
α	1
β	2
γ	3

r \cap s

A	B
β	2



Difference

$$r - s$$

r

A	B
α	3
β	2

s

A	B
α	1
β	2
γ	3

$r - s$

A	B
α	3

Can you find an expression that is equivalent to $r - s$ using the operators $\cap$ and $\sim$?



Selection

r

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

$\sigma_{(A=B) \wedge (D>5)} (r)$

A	B	C	D
α	α	1	7
β	β	23	10



Projection

r

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3
β	β	23	10

$\pi_{A,C}(r)$

A	C
α	1
α	5
β	12
β	23



Join

Cartesian Product

r

A	B
α	3
β	2

s

C	D
α	1
β	2
γ	3

r $\times$ *s*

A	B	C	D
α	3	α	1
α	3	β	2
α	3	γ	3
β	2	α	1
β	2	β	2
β	2	γ	3



Join

Cartesian Product

r

A	B
α	3
β	2

s

A	C
α	1
β	2
γ	3

r $\times$ *s*

r.A	B	s.A	C
α	3	α	1
α	3	β	2
α	3	γ	3
β	2	α	1
β	2	β	2
β	2	γ	3



Rename

Rename operation $\rho_x(E)$ returns the output of the expression E under the name x

r

A	B
α	3
β	2

$r \times \rho_s(r)$

r.A	r.B	s.A	s.B
α	3	α	3
α	3	β	2
β	2	α	3
β	2	β	2



Natural Join

- Let r and s be relations on schemas R and S , respectively.
- Natural join of relations R and S is a relation on schema $R \cup S$ obtained as follows:
 - Consider each pair of tuples t_r from r and t_s from s .
 - If t_r and t_s have the same value on each of the attributes in $R \cap S$, add a tuple t to the result, where
 - t has the same value as t_r on r
 - t has the same value as t_s on s



Natural Join – Example

r

A	B	C	D
α	1	α	a
β	2	γ	a
γ	4	β	b
α	1	γ	a
δ	2	β	b

s

B	D	E
1	a	α
3	a	β
1	a	γ
2	b	δ
3	b	ϵ

$r \bowtie s$

A	B	C	D	E
α	1	α	a	α
α	1	α	a	γ
α	1	γ	a	α
α	1	γ	a	γ
δ	2	β	b	δ

$$r \bowtie s \equiv \pi_{A,r.B,C,r.D,E}(\sigma_{r.B=s.B \wedge r.D=s.D} (r \times s))$$



Composition of Operations

r

A	B
α	3
β	2

s

A	B
α	1
β	2
γ	3

$r \times s$

r.A	r.B	s.A	s.B
α	3	α	1
α	3	β	2
α	3	γ	3
β	2	α	1
β	2	β	2
β	2	γ	3

$\sigma_{r.A=s.A}(r \times s)$

r.A	r.B	s.A	s.B
α	3	α	1
β	2	β	2



Summary of Relational Algebra Operators

Symbol (name)	Description
σ (Selection)	Return rows of the input relation that satisfy a predicate.
π (Projection)	Return specified attributes from all rows of the input relation. Remove duplicate tuples from the output.
$\times$ (Cartesian Product)	Return pairs of rows from the <i>two</i> input relations.
$\cup$ (Union)	Return the union of tuples from the <i>two</i> input relations.
$-$ (Difference)	Return the set of records that exist in the relation before the operator(-) but not in the relation after the operator.
$\cap$ (Intersection)	Return the common tuples in both input relations.
$\bowtie$ (Natural Join)	Return pairs of rows from the <i>two</i> input relations that have the same value on all attributes that have the same name.
ρ (Rename)	Returns the outcome of an expression under the specified name



Remarks on RA

- Each Query input is a table (or set of tables)
- Each query output is a table.
- All data in the output table appears in one of the input tables
- Can we compute:
 - SUM $\gamma_{SUM}(tot_credit)(student)$
 - AVG $\gamma_{AVG}(salary)(instructor)$
 - MAX $\gamma_{MAX}(budget) \rightarrow max_budget(department)$
 - MIN $\gamma_{MIN}(budget) \rightarrow min_budget(department)$
 - COUNT $\gamma_{COUNT}(tot_cred \geq 12) \rightarrow high_cred(student)$
 - GROUP BY $dept_name \gamma_{AVG}(salary) \rightarrow dept_avg_salary(instructor)$



Data Extraction Using Structured Query Language (SQL)



SELECT-FROM-WHERE Statements

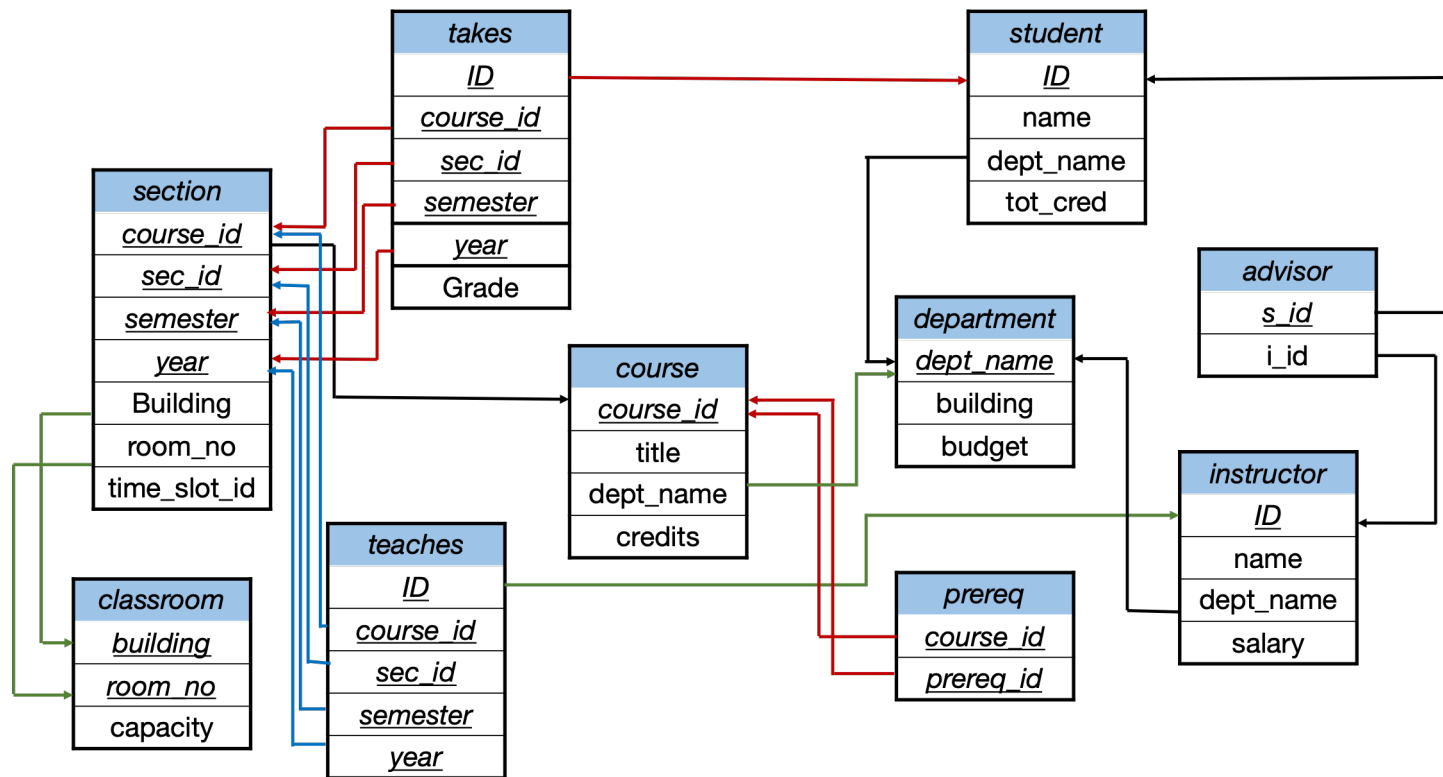
SELECT desired attributes

FROM one or more tables

WHERE condition about tuples of the tables



Remember Our Main Database?



Example: The SELECT Clause

- Get the IDs, names and total credits of students who completed at least 24 credits

```
SELECT ID, name, tot_cred  
FROM student  
WHERE tot_cred >= 24;
```

- The answer is a relation with three attributes (ID, name, tot_cred)
- What if we need only the names of the students?
- Note that: the SQL operator names are case insensitive SELECT $\equiv$ Select $\equiv$ select



The SELECT Clause

- Lists the desired attributes in the result of the query
 - Corresponds to the projection operator of the relational algebra
- SQL allows duplicates in relations as well as in query results
- To force the elimination of duplicates, use the keyword **DISTINCT** after select
- Example: find the department names of all instructors whose salary is strictly greater than 60000 without showing the department name more than once

```
SELECT DISTINCT dept_name  
FROM instructor  
WHERE salary > 60000
```



The SELECT Clause (Cont.)

- The keyword **ALL** specifies that duplicates should not be removed

```
SELECT ALL dept_name  
FROM instructor  
WHERE salary > 60000
```

- An asterisk in the **SELECT** clause denotes “all attributes”

```
SELECT * FROM instructor
```

will return all the records from table “instructor”



The SELECT Clause (Cont.)

- An attribute could be a literal with no **FROM** clause

SELECT '542'

Results in a relation with one column and one row with value "542"

We can also give the column a name using

SELECT '542' **AS** V1

- An attribute could be a literal with **FROM** clause

SELECT 'A' **FROM** instructor

will return a relation with one column and N rows (the number of tuples in the 'instructor' relation) where each row will contain the value "A"



The SELECT Clause (Cont.)

- **SELECT** clause can contain arithmetic expressions involving the operators *, +, -, and /.

```
SELECT ID, name, salary/12.0
```

```
FROM instructor
```

This query would return a relation with the same number of records as the 'instructor' relation and the (ID, name, salary/12.0) where the yearly salary is replaced by the monthly salary

We can rename the "salary/12.0" using the **AS** clause

```
SELECT ID, name, salary/12.0 as monthly_salary
```

```
FROM instructor
```



The WHERE Clause

- Specifies conditions that the result must satisfy
 - Corresponds to the selection predicate of the relational algebra
- To find all students enrolled in the 'Math.' department

```
SELECT ID, name FROM student  
WHERE dept_name = 'Math.'
```

- Conditions can be also combined using logical operators (AND, OR, NOT)
 - Find all students in the 'Math.' department who completed a minimum of 24 credits

```
SELECT ID, name FROM student  
WHERE dept_name = 'Math.' AND tot_cred >= 24
```

- Comparisons =, <>, <, >, <=, >=



The FROM Clause

- Lists the relations involved in the query
 - Corresponds to the Cartesian product operation of the relational algebra
- Find the Cartesian product '*instructor*' X '*teaches*'

SELECT * **FROM** instructor, teaches

- Generates every possible instructor–teaches pairs, with all attributes from both relations.
- For common attributes (e.g., *ID*), the attributes in the resulting table are renamed using the relation name (e.g., *instructor.ID*)
- Similar to the cartesian product in RA
- Cartesian product not very useful directly, but useful when combined with where-clause condition.



SELECT-FROM-WHERE Examples

```
SELECT name, course_id  
FROM instructor, teaches  
WHERE instructor.ID = teaches.ID
```

Returns the names of all instructors who have taught any courses and the course_id

```
SELECT name, course_id  
FROM instructor, teaches  
WHERE instructor.ID = teaches.ID AND instructor.dept_name = 'Art'
```

Returns the names of all instructors in the Art department who have taught any courses and the course_id



The Rename Operation

- SQL allows renaming relations and attributes using the **AS** clause:

old-name **AS** *new-name*

- Find the names of all instructors who have a higher salary than some instructor in 'Comp. Sci'.

```
SELECT DISTINCT T.name  
FROM instructor AS T, instructor AS S  
WHERE T.salary >= 75000 AND S.dept_name = 'Comp. Sci.'
```

Returns the names of instructors in the 'Comp. Sci.' department joined with whose salary is greater than or equal 75000

- Keyword **AS** is optional and may be omitted
instructor **AS** *T* $\equiv$ *instructor T*



Renaming Example: Self-Join

- Sometimes, a query needs to use two copies of the same relation.
- Distinguish copies by renaming the relations.

- Example self-Join

```
SELECT T.name AS N1, S.name AS N2  
FROM instructor T, instructor S  
WHERE T.salary = S.salary AND T.name < S.name
```

- Returns the names of instructors who have the same salary
 - Do not produce pairs like (Miller, Miller)
 - Produces pairs in alphabetic order, e.g. (Addison, Miller), not (Miller, Addison)
- Note that we omit AS when renaming the relations



Join Operation

- **JOIN** operations take two relations and return as a result another relation.
- A join operation is a Cartesian product which requires that tuples in the two relations match (under some condition). It also specifies the attributes that are present in the result of the join
- The join operations are typically used as subquery expressions in the **FROM** clause



Join Operation (Cont.)

- We will consider the following relations in the few coming slides
- Relation *course*

course_id	title	dept_name	credits
BIO-301	Genetics	Biology	4
CS-490	Game Design	Comp. Sci.	4
CS-315	Boolean Algebra	Comp. Sci.	3

- Relation *prereq*

course_id	prereq_id
BIO-301	BIO-101
CS-490	CS-101
CS-347	CS-201

- Note that:

- *prereq* information is missing for course CS-315
- *course* information is missing for course CS-347



Outer Join

- An extension of the join operation that avoids loss of information.
- Computes the join and then adds tuples from one relation that does not match tuples in the other relation to the result of the join.
- Uses *null* values.



Left Outer Join

```
SELECT *  
FROM course  
LEFT OUTER JOIN prereq  
ON course.course_id = prereq.course_id
```

course_id	title	dept_name	credits	prereq_id
BIO-301	Genetics	Biology	4	BIO-101
CS-490	Game Design	Comp. Sci.	4	CS-101
CS-315	Boolean Algebra	Comp. Sci.	3	<i>null</i>



Right Outer Join

```
SELECT *  
FROM course  
RIGHT OUTER JOIN prereq  
ON course.course_id = prereq.course_id
```

course_id	prereq_id	title	dept_name	credits
BIO-301	BIO-101	Genetics	Biology	4
CS-490	CS-101	Game Design	Comp. Sci.	4
CS-347	CS-201	<i>null</i>	<i>null</i>	<i>null</i>

- Remember:
 - The order of the attributes in a relation has no meaning



Full Outer Join

```
SELECT *  
FROM course  
FULL OUTER JOIN prereq  
ON course.course_id = prereq.course_id
```

course_id	prereq_id	title	dept_name	credits
BIO-301	BIO-101	Genetics	Biology	4
CS-490	CS-101	Game Design	Comp. Sci.	4
CS-347	CS-201	<i>null</i>	<i>null</i>	<i>null</i>
CS-315	<i>null</i>	Boolean Algebra	Comp. Sci.	3



Inner Join

```
SELECT *  
FROM course  
INNER JOIN prereq  
ON course.course_id = prereq.course_id
```

course_id	prereq_id	title	dept_name	credits
BIO-301	BIO-101	Genetics	Biology	4
CS-490	CS-101	Game Design	Comp. Sci.	4

- An SQL INNER JOIN is same as JOIN clause
- Question:
 - What is the difference between the above JOIN and the right/left outer join



String Operations

- SQL includes a string-matching operator for comparisons on character strings.
- The operator **LIKE** uses patterns that are described using two special characters
 - Percent (%). The % character matches any substring.
 - Underscore (_). The _ character matches any character.
- Example: find the names of all instructors whose name includes the substring “Van der”

```
SELECT DISTINCT name  
FROM instructor WHERE name LIKE '%Van der%'
```



String Operations (Cont.)

- Match the string “100%”

LIKE ‘100 \%' ESCAPE ‘\’

in that above we use backslash (\) as the escape character

- Patterns are case sensitive.
- Pattern matching examples:
 - ‘Intro%’ matches any string beginning with “Intro”.
 - ‘%Comp%’ matches any string containing “Comp” as a substring.
 - ‘___’ matches any string of exactly three characters.
 - ‘___ %’ matches any string of at least three characters.



Range Queries

- SQL includes a **BETWEEN** comparison operator
- Example: Find the names of all instructors with salary between \$90,000 and \$100,000 (that is, $\geq$ \$90,000 and $\leq$ \$100,000)

```
SELECT name  
FROM instructor  
WHERE salary BETWEEN 90000 AND 100000
```



Tuple Comparison

```
SELECT name, course_id  
FROM instructor, teaches  
WHERE (instructor.ID, dept_name) = (teaches.ID, 'Biology')
```



Set Operations

- Find courses that ran in Fall 2009 or in Spring 2010
`SELECT course_id FROM section WHERE sem = 'Fall' AND year = 2009`
`UNION`
`SELECT course_id FROM section WHERE sem = 'Spring' AND year = 2010`
- Find courses that ran in Fall 2009 and in Spring 2010
`SELECT course_id FROM section WHERE sem = 'Fall' AND year = 2009`
`INTERSECT`
`SELECT course_id FROM section WHERE sem = 'Spring' AND year = 2010`
- Find courses that ran in Fall 2009 but not in Spring 2010
`SELECT course_id FROM section WHERE sem = 'Fall' AND year = 2009`
`EXCEPT`
`SELECT course_id FROM section WHERE sem = 'Spring' AND year = 2010`



Null Values

- It is possible for tuples to have a null value, denoted by *NULL*, for some of their attributes
- *null* signifies an unknown value or that a value does not exist.
- The result of any arithmetic expression involving *NULL* is *NULL*
 - Example: $5 + NULL$ returns *NULL*
- The predicate **IS NULL** can be used to check for null values.
 - Example: Find all instructors whose salary is null.

```
SELECT name
FROM instructor
WHERE salary IS NULL
```



Null Values and Three Valued Logic

- Three values – *true*, *false*, *unknown*
- Any comparison with *null* returns *unknown*
 - Example: $5 < \text{null}$ or $\text{null} <> \text{null}$ or $\text{null} = \text{null}$
- Three-valued logic using the value *unknown*:
 - OR: (*unknown* **OR** *true*) = *true*,
(*unknown* **OR** *false*) = *unknown*
(*unknown* **OR** *unknown*) = *unknown*
 - AND: (*true* **AND** *unknown*) = *unknown*,
(*false* **AND** *unknown*) = *false*,
(*unknown* **AND** *unknown*) = *unknown*
 - NOT: (**NOT** *unknown*) = *unknown*
 - “*P* is **unknown**” evaluates to *true* if predicate *P* evaluates to *unknown*
- Result of **WHERE** clause predicate is treated as *false* if it evaluates to *unknown*

$\wedge$	<i>T</i>	<i>F</i>	<i>U</i>
<i>T</i>	T	F	U
<i>F</i>	F	F	F
<i>U</i>	U	F	U

$\vee$	<i>T</i>	<i>F</i>	<i>U</i>
<i>T</i>	T	T	T
<i>F</i>	T	F	U
<i>U</i>	T	U	U



The IN Operator

- `<v> IN <S>` evaluates to true if the value `v` matches one of the values in `S`.
- It can be used to replace a sequence of conditions connected by **OR**
- **Example:**

```
SELECT name  
FROM instructor  
WHERE dept_name IN ('Comp. Sci.', 'Math.', 'Chem.');
```

This Query is equivalent to:

```
SELECT name  
FROM instructor  
WHERE dept_name = 'Comp. Sci.' OR dept_name = 'Math.' OR dept_name =  
'Chem.'
```





Wrap-Up

- Summarize what you learned today in 2-minutes





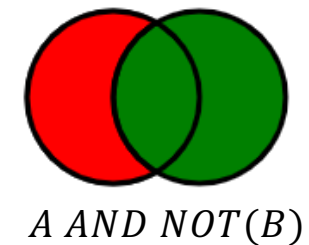
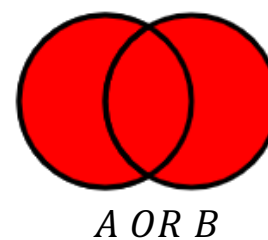
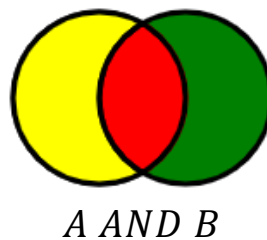
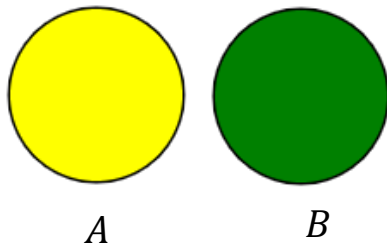
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but no rights can be derived from its contents.

Boolean Operators – Revision



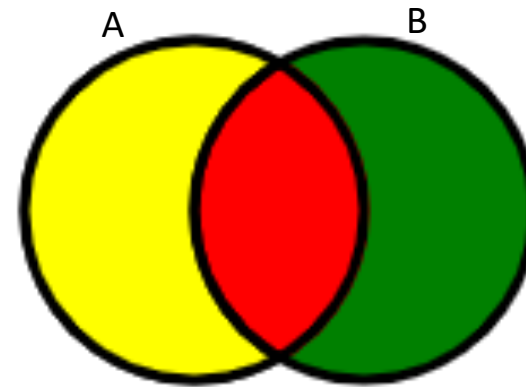
Boolean Operators

- Searching through a database or search engine can often be frustrating
- Boolean Operators create relationships between concepts and terms for better search results
- Most popular Boolean operators are **AND**, **OR** and **NOT**
 - The red areas represent the results of the operators



AND ($\wedge$)

A	B	$A \wedge B$
1	1	1
1	0	0
0	1	0
0	0	0

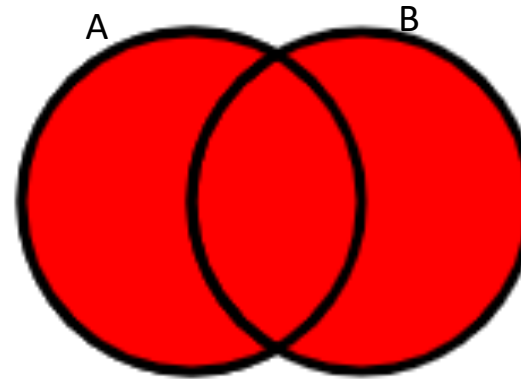


The red area – ($A \wedge B$)

- Retrieves only records that satisfy both conditions
- Example:
name = “Taylor” AND dept_name = “Chem.”
Returns all instructors in the Chem. department whose name is Taylor

OR ($\vee$)

A	B	$A \vee B$
1	1	1
1	0	1
0	1	1
0	0	0



The red area – $(A \vee B)$

- Retrieves records that satisfy one of the conditions

- Example:

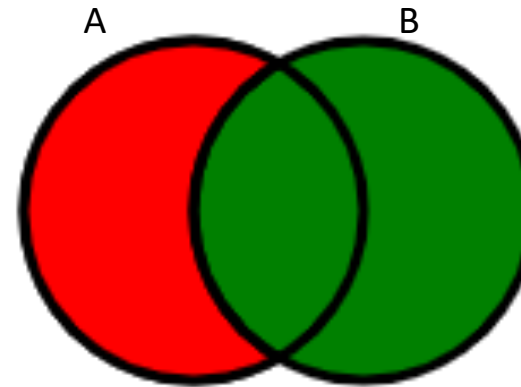
name = "Taylor" OR dept_name = "Chem."

Returns all instructors with the name Taylor and all instructors of the Chem. department



NOT ($\sim$)

A	$\sim A$
1	0
0	1



The red area – $A \text{ AND } \text{NOT}(B)$

- Retrieves only records that satisfy the first condition and doesn't satisfy the second
- Example:
name = "Taylor" AND
NOT dept_name = "Chem."
Returns all instructors with the name Taylor who do not work in the Chem. department



Boolean Equivalence

- Equivalence of two Boolean operations can be easily proven using truth tables
- Equivalence of Boolean operations is useful for optimizing the Boolean queries
- Examples:

- $\sim (A \wedge B) \equiv \sim A \vee \sim B \longrightarrow$

<i>A</i>	<i>B</i>	$(A \wedge B)$	$\sim (A \wedge B)$	$\sim A \vee \sim B$
1	1	1	0	0
1	0	0	1	1
0	1	0	1	1
0	0	0	1	1

$\equiv$

- $(A \wedge B) \wedge \sim B \equiv A \wedge (B \wedge \sim B) \equiv \text{false}$
 - $(A \wedge B) \vee (\sim B \wedge C) \vee (A \wedge C) \vee \sim (A \wedge C) \equiv (A \wedge C) \vee \sim (A \wedge C) \equiv \text{true}$



Proving Boolean Equivalence

- Truth Table: Helpful when the number of statements in the Boolean expression is small
- Proof by contradiction: assume the expression is false/true and proof that it leads to contradiction
- Using the Boolean axioms.



Boolean Axioms

- Let T and F represent the cases of always True and always False
 - $(T \wedge T) = T$
 - $(F \wedge F) = F$
 - $(T \vee T) = T$
 - $(F \vee F) = F$
 - $(T \wedge F) = (F \wedge T) = F$
 - $(T \vee F) = (F \vee T) = T$
 - $\overline{T} = F$
 - $\overline{F} = T$



Boolean Axioms

- Commutativity
 - $(A \wedge B) = (B \wedge A)$
 - $(A \vee B) = (B \vee A)$
- Identity
 - $(A \wedge T) = A$
 - $(A \vee T) = T$
 - $(A \wedge F) = F$
 - $(A \vee F) = A$



Boolean Axioms

- Idempotent
 - $(A \wedge A) = A$
 - $(A \vee A) = A$
- Involution
 - $\overline{\overline{A}} = A$
- Complement
 - $(A \wedge \overline{A}) = F$
 - $(A \vee \overline{A}) = T$



Boolean Axioms

- Associativity
 - $(A \wedge B) \wedge C = A \wedge (B \wedge C)$
 - $(A \vee B) \vee C = A \vee (B \vee C)$
- Distributivity
 - $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$
 - $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$



Proving Boolean Equivalence

- Using the Boolean axioms (Example):

- prove that: $(A \wedge B) \vee (\sim A \wedge C) \vee (B \wedge C) = (A \wedge B) \vee (\sim A \wedge C)$

- $$\begin{aligned} \text{LHS} &= (A \wedge B) \vee (\sim A \wedge C) \vee (B \wedge C) = (A \wedge B) \vee (\sim A \wedge C) \vee [(B \wedge C) \wedge \mathbf{T}] \\ &= (A \wedge B) \vee (\sim A \wedge C) \vee [(B \wedge C) \wedge (A \vee \sim A)] \\ &= (A \wedge B) \vee (\sim A \wedge C) \vee [(B \wedge C \wedge A) \vee (B \wedge C \wedge \sim A)] \\ &= [(A \wedge B) \vee (B \wedge C \wedge A)] \vee [(\sim A \wedge C) \vee (B \wedge C \wedge \sim A)] \\ &= [(A \wedge B) \wedge (\mathbf{T} \vee C)] \vee [(\sim A \wedge C) \wedge (\mathbf{T} \vee B)] \\ &= [(A \wedge B)] \vee [(\sim A \wedge C)] = \text{RHS} \end{aligned}$$

- $\mathbf{T}$ means always TRUE.



Read more FROM
John Kelly, The Essence of Logic (Prentice Hall, 1997)
Chapter 1 is good enough

